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4R/2025 MAT4204C

2025

MATHEMATICS

(Major-II)

Paper : MAT4204C

(Complex Analysis)

Full Marks : 45

Time : Two hours

***The figures in the margin indicate
full marks for the questions.***

1. Answer all the questions : $1 \times 4 = 4$

(a) What is the principal value $(i)^i$?

(b) Does $\lim_{z \rightarrow 0} \frac{z}{z}$ exist ? Justify.

(c) What is the n th term of the sequence

$$\frac{2}{3!}, \frac{4}{5!}, \frac{6}{7!}, \dots ?$$

- (d) Evaluate $\oint_C \frac{z^2}{z-1} dz$, where C is any simple closed curve containing $z = 1$ inside.

2. Answer **all** the questions : $2 \times 3 = 6$

- (a) Find the image of the region $1 \leq |z| \leq 2$ under the mapping $w = z^2$.

- (b) Without actually evaluating the integral

show that $\left| \int_C \frac{dz}{z^2 + 1} \right| \leq \frac{d\pi}{3}$, where C is the

arc of the circle $|z| = 2$ from $z = 2$ to $z = -2$ and lies in the upper half-plane.

- (c) Find the radius of convergence of the

series $x + \frac{x^2}{2^2} + \frac{x^3}{2!3^3} + \frac{x^4}{3!4^4} + \dots$

3. Answer the following questions : $5 \times 3 = 15$

- (a) Let $w(t)$ be a piecewise continuous complex valued function defined in an interval $a \leq t \leq b$, then show that

$$\left| \int_a^b w(t) dt \right| \leq \int_a^b |w(t)| dt.$$

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- (b) Define a harmonic function. Prove that $u(x, y) = e^x \cos y$ is harmonic. Find a harmonic conjugate of u .

OR

- (c) Show that $\sin z = \sin x \cosh y + i \cos x \sinh y$.

- (d) Show that

$$\int_C (z - z_0)^m dz = \begin{cases} 2\pi i, & m = -1 \\ 0, & m \neq -1 \text{ and integer,} \end{cases}$$

where C is any circle centred at z_0 .

OR

- (e) Find the series expansion of $f(z) = \frac{1}{z}$ about $z = 1$.

4. Answer **either** (a) **or** (b) :

- (a) Let $f(z) = u(x, y) + iv(x, y)$ be differentiable at a point $z = x + iy$. Show that the first order partial derivatives of $u(x, y)$ and $v(x, y)$

satisfy $u_x(x, y) = v_y(x, y)$ and

$$u_y(x, y) = -v_x(x, y).$$

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Contd.

- (b) Suppose $f(z)$ is analytic in a region D and is not identically zero in D , then show that the set of all zeros of $f(z)$ is isolated. 10

5. Answer **either** (a) **or** (b) :

(a) State and prove the Cauchy's integral

formula. Use it to evaluate $\oint_C \frac{e^z}{(z-3)^2} dz$,

where C is the circle $|z-3|=1$. 10

(b) State and prove the Fundamental Theorem of Algebra. 10

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+1}}$$

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{(n-1)! n^n}}{1}$$

$$\frac{x^n}{(n-1)! n^n} \quad \frac{x}{0! 1!} + \frac{x^2}{1! 2!} + \frac{x^3}{2! 3!}$$

$$\lim_{n \rightarrow \infty} \frac{x^n}{(n-1)! n^n}$$

$$\frac{2 \times 1}{3!}, \frac{2 \times 2}{5!}, \frac{2 \times 3}{7!}$$

$$\frac{2n}{(n+1)!}$$

$$171 = n+4$$

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$$\frac{n+1}{n+1}$$

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